

ALEXANDER J. TURNER

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EDUCATION

Ph.D., Harvard University Atmospheric Chemistry, Advisor: Daniel J. Jacob	2012 – 2017
B.S., University of Colorado at Boulder Mechanical Engineering, Advisor: Daven K. Henze	2008 – 2012

PROFESSIONAL EXPERIENCE

Associate Professor , University of Washington	2026 –
Assistant Professor , University of Washington	2021 – 2026
Calvin Professor of Atmospheric Science , University of Washington	2022 – 2024
Miller Postdoctoral Fellow , University of California at Berkeley	2017 – 2020
Graduate Research Assistant , Harvard University	2012 – 2017

AWARDS AND FELLOWSHIPS

AGU James B. Macelwane Medal	2026
Fellow of the American Geophysical Union	2026
UW Atmospheric and Climate Science Annual Teaching Award	2022, 2023, 2026
AGU James R. Holton Award	2020
Miller Fellowship at UC Berkeley	2017 – 2020
DOE Computational Science Graduate Fellowship (CSGF)	2013 – 2017
CU Boulder College of Engineering's "Outstanding Graduate for Research"	2012
NOAA Ernest F. Hollings Scholar	2010 – 2012

PUBLICATIONS (GROUP MEMBER, *SUBMITTED, #CONTRIBUTED EQUALLY)

h-index = 31, total citations = 4,777 (as of September 23, 2026: scholar.google.com/citations?user=tfmVgv0AAAAAJ)

- *65. Yoon, J. Y. S., J. A. Thornton, P. N. Blossey, M. F. Khairoutdinov, M. C. Wyant, J. Kesselmeier, S. Kim, C. M. Nussbaumer, C. Sarkar, R. Seco, J. Shilling, J. N. Smith, A. M. Yanez-Serrano, and **A. J. Turner** (submitted), gSAM-Chem: Development and Evaluation of a Large Eddy Simulation with Interactive Chemistry, *submitted*.
- *64. Ruhm, R., E. Chaboteaux, E. G. Cosio, R. S. Cruz Chino, S. E. Janssen, S. A. Plunkett, E. Ramos Chang, M. T. Tate, **A. J. Turner**, and R. B. Neumann (submitted), A high-resolution approach to estimate gold-mining mercury emissions and their downwind concentration footprints, *submitted*.
- *63. Dadheech, N. and **A. J. Turner** (submitted), Observing methane emissions from oil and gas infrastructure in the Permian Basin using TROPOMI, *submitted*.
- *62. Manchanda, C.[#], L. H. Koolik[#], A. Unal, I. Fung, J. D. Marshall, R. Morello-Frosch, **A. J. Turner**, R. A. Harley, J. S. Apte (submitted), Inverse modeling identifies efficient emission control strategy for mitigating PM_{2.5} pollution, *submitted*.
- *61. Nassau, R., L. Jaeglé, and **A. J. Turner** (submitted), Observed stagnation in United States black carbon reveals overestimated emission declines, *submitted*.
- *60. Manchanda, C., R. C. Cohen, R. A. Alvarez, T. M. Thompson, M. H. Harris, **A. J. Turner**, J. D. Marshall, R. A. Harley, and J. S. Apte (submitted), Hyperlocal sensing and inversion reveal community impacts of urban air pollutant emissions, *submitted*.
- 59. Ort, L., C. Frankenberg, N. Dadheech, J. Y. S. Yoon, **A. J. Turner**, A. A. Bloom, P. Konopka, H. C. Lachnitt, P. Hoor, J. Lelieveld, E. A. Kort, and H. Fischer (2026), Airborne observations reveal underestimated riverine methane emissions across the Amazon, *Geophys. Res. Lett.*, 53, e2026GL122310, doi:10.1029/2026GL122310.
- 58. Mei, E. J., G. J. Hakim, C. Proistosescu, T. K. Bauska, C. Buizert, and **A. J. Turner** (2026), Multidecadal preindustrial methane variability can be explained by noise in the source-sink imbalance, *Proc. Natl. Acad. Sci.*, 123 (25), doi:10.1073/pnas.2601235123.

57. Rao, M. P., A. Pacheco-Solana, R. Li, B. Oryan, J. E. Jensen, M. Rodriguez-Caton, L. Klinek, Z. A. Pierrat, S. Ruehr, R. Oelkers, L. E. Boeschoten, K. L. Griffin, M. L. McCormack, X. Yang, J. Verfaillie, D. Baldocchi, J. Hise, **A. J. Turner**, T. M. Scanlon, L. Andreu-Hayles, J. U. H. Eitel, N. Pederson, D. Griffin, D. Stahle, J. T. Maxwell, S. Voelker, S. A. Kannenberg, J. Peñuelas, and T. S. Magney (2026), Decoupled carbon assimilation and growth responses to aridity in temperate deciduous oaks, *Sci. Adv.*, *12*, eady7139, doi:10.1126/sciadv.ady7139.
56. Balasus, N., D. J. Jacob, A. A. Bloom, J. D. East, L. A. Estrada, S. E. Hancock, M. He, T. A. Mooring, **A. J. Turner**, and J. R. Worden (2026), 2019–2024 trends in African livestock and wetland emissions as contributors to the global methane rise, *Atmos. Chem. Phys.*, *26*, 4601–4617, doi:10.5194/acp-26-4601-2026.
55. Yoon, J. Y. S., K. Wells, D. B. Millet, C. Frankenberg, S. Sanghavi, A. L. S. Swann, J. A. Thornton, and **A. J. Turner** (2026), Inferring drivers of tropical isoprene: competing effects of emissions and chemistry, *Atmos. Chem. Phys.*, *26*, 4509–4529, doi:10.5194/acp-26-4509-2026.
54. Dadheech, N. and **A. J. Turner** (2026), Simulating out-of-sample atmospheric transport to enable flux inversions, *Atmos. Chem. Phys.*, *26*, 427–441, doi:10.5194/acp-26-427-2026.
53. Liu, A. X., C. M. Zarakas, B. G. Buchovecky, L. R. Hawkins, A. Cordak, A. Cornish, M. Haagsma, G. J. Kooperman, C. J. Still, C. D. Koven, **A. J. Turner**, D. S. Battisti, J. T. Randerson, F. M. Hoffman, and A. L. S. Swann (2025), Photosynthesis responses to intrinsic water use efficiency depend on atmospheric feedbacks and modify the magnitude of response to elevated CO₂, *J. Adv. Model. Earth Syst.*, *17*, e2025MS005177, doi:10.1029/2025MS005177.
52. Mei, E. J., G. J. Hakim, M. Taniguchi-King, D. Stiller, and **A. J. Turner** (2025), Emulating chemistry-climate dynamics with a linear inverse model, *Atmos. Chem. Phys.*, *25*, 15033–15045, doi:10.5194/acp-25-15033-2025.
51. Johnson, M. S., S. D. Hamilton, S. Jeong, Y. Cui, D. Wu, **A. J. Turner**, and M. L. Fischer (2025), State-wide California 2020 Carbon Dioxide Budget Estimated with OCO-2 and OCO-3 satellite data, *Atmos. Chem. Phys.*, *25*, 8475–8492, doi:10.5194/acp-25-8475-2025.
50. Yoon, J. Y. S., K. Wells, D. B. Millet, A. L. S. Swann, J. A. Thornton, and **A. J. Turner** (2025), Impacts of interannual isoprene variations on methane lifetimes and trends, *Geophys. Res. Lett.*, *52*, e2025GL114712, doi:10.1029/2025GL114712.
49. Asimow, N. G., M. Y. Patel, Y. Zhu, A. R. Winter, K. R. Gurney, W. M. Berelson, **A. J. Turner**, and R. C. Cohen (2025), Differences in regional home heating behaviour in three U.S. cities revealed by ground-based sensor network, *Geophys. Res. Lett.*, *52*, e2025GL115772, doi:10.1029/2025GL115772.
48. Dadheech, N.[#], T. He[#], and **A. J. Turner** (2025), High-resolution greenhouse gas flux inversions using a machine learning surrogate model for atmospheric transport, *Atmos. Chem. Phys.*, *25*, 5159–5174, doi:10.5194/acp-25-5159-2025.
47. Liu, G., L. Shen, P. Ciais, X. Lin, D. Hauglustaine, X. Yi, X. Lan, **A. J. Turner**, and S. Peng (2025), Increased Arctic natural emissions and tropical tropospheric hydroxyl radicals explain changes in seasonal amplitude of atmospheric CH₄, *Nature*, *641*, 660–665, doi:10.1038/s41586-025-08900-8.
46. He, T.[#], G. M. Oomen[#], W. Tang, I. Bouarar, K. Chance, C. Clerbaux, D. Edwards, H. Eskes, B. Gaubert, C. Granier, M. Guevara, D. Jacob, J. Kaiser, J. Kim, S. Kondragunta, X. Liu, E. Marais, K. Miyazaki, R. Park, V. H. Peuch, G. Pfister, A. Richter, T. Stavrou, R. M. Suleiman, **A. J. Turner**, B. Veihelmann, Z. C. Zeng, and G. Brasseur (2025), Challenges and opportunities offered by geostationary space observations for air quality research and emission monitoring, *Bull. Am. Meteorol. Soc.*, *106*, E939–E963, doi:10.1175/BAMS-D-23-0145.1.
45. He, T.[#], N. Dadheech[#], T. M. Thompson, and **A. J. Turner** (2025), FootNet v1.0: Development of a machine learning emulator of atmospheric transport, *Geosci. Model Dev.*, *18*, 1661–1671, doi:10.5194/gmd-18-1661-2025.
44. Jeong, S., S. D. Hamilton, D. Wu, M. S. Johnson, **A. J. Turner**, and M. L. Fischer (2025), Applying Gaussian Process Machine Learning and Modern Probabilistic Programming to Satellite Data to Infer CO₂ Emissions, *Environ. Sci. Technol.*, *59*, 4376–4387, doi:10.1021/acs.est.4c09395.
43. Hamilton, S. D., D. Wu, M. S. Johnson, **A. J. Turner**, M. L. Fischer, N. Dadheech, and S. Jeong (2024), Estimating carbon dioxide emissions in two California cities using Bayesian inversion and satellite measurements, *Geophys. Res. Lett.*, *e2024GL111150*, doi:10.1029/2024GL111150.
42. Dressel, I., S. Zhang, M. A. G. Demetillo, S. Yu, K. Fields, L. M. Judd, C. R. Nowlan, K. Sun, A. Kotsakis, **A. J. Turner**, and S. E. Pusede (2024), Neighborhood-level nitrogen dioxide inequalities contribute to surface ozone in Houston, Texas, *Environ. Sci. Technol. Air*, *1*, 973–988, doi:10.1021/acsestair.4c00009.

41. Manchanda, C., R. Harley, J. D. Marshall, **A. J. Turner**, and J. S. Apte (2024), Integrating Mobile and Fixed-Site Black Carbon Measurements to Bridge Spatiotemporal Gaps in Urban Air Quality, *Environ. Sci. Technol.*, 58, 12563–12574, doi:10.1021/acs.est.3c10829.
40. Lee, B. H., J. M. Munger, S. C. Wofsy, L. V. Rizzo, J. Y. S. Yoon, **A. J. Turner**, J. A. Thornton, and A. L. S. Swann (2024), Sensitive response of atmospheric oxidative capacity to the uncertainty in the emissions of nitric oxide (NO) from soils in Amazonia, *Geophys. Res. Lett.*, e2023GL107214, doi:10.1029/2023GL107214.
39. Asimow, N. G., **A. J. Turner**, and R. C. Cohen (2024), Sustained reductions of Bay Area CO₂ emissions 2018–2022, *Environ. Sci. Technol.*, 58, 6586–6594, doi:10.1021/acs.est.3c09642.
38. He, T., R. J. Boyd, D. J. Varon, and **A. J. Turner** (2024), Increased methane emissions from oil and gas following the Soviet Union's collapse, *Proc. Natl. Acad. Sci.*, 121, doi:10.1073/pnas.2314600121.
37. Moon, A., U. Jongebloed, K. K. Dingilian, A. J. Schauer, Y. C. Chan, M. Cesler-Maloney, W. R. Simpson, R. J. Weber, L. Tsiang, F. Yazbeck, S. Zhai, A. Wedum, **A. J. Turner**, S. Albertin, S. Bekki, J. Savarino, K. Gribanov, K. A. Pratt, E. J. Costa, C. Anastasio, M. O. Sunday, L. M. D. Heinlein, J. Mao, and B. Alexander (2023), Primary Sulfate Is the Dominant Source of Particulate Sulfate During Winter in Fairbanks, Alaska, *Environ. Sci. Technol. Air*, 1, 139–149, doi:10.1021/acsestair.3c00023.
36. Yu, X., D. B. Millet, D. K. Henze, **A. J. Turner**, A. L. Delgado, A. A. Bloom, and J. Sheng (2023), A high-resolution satellite-based map of global methane emissions reveals missing wetland, fossil fuel, and monsoon sources, *Atmos. Chem. Phys.*, 23, 3325–3346, doi:10.5194/acp-23-3325-2023.
35. Hajny, K. D., C. Floerchinger, I. Lopez-Coto, J. Pitt, C. Gately, K. Gurney, L. Hutyrá, T. Jayarathne, R. Kaeser, G. Roest, M. Sargent, B. H. Stirm, J. Tomlin, **A. J. Turner**, P. B. Shepson, and S. Wofsy (2022), Measurements of anthropogenic CO₂ emissions from New York City compared to inventories, *Elem. Sci. Anth.*, 10:1, doi:10.1525/elementa.2021.00121.
34. Kim, J., **A. J. Turner**, H. Fitzmaurice, E. Delaria, C. Newman, P. J. Wooldridge, and R. C. Cohen (2022), Observing annual trends in vehicular CO₂ emissions, *Environ. Sci. Technol.*, 56, 3925–3931, doi:10.1021/acs.est.1c06828.
33. Gensheimer, J., **A. J. Turner**, P. Köhler, C. Frankenberg, and J. Chen (2022), A convolutional neural network for spatial downscaling of satellite-based solar-induced chlorophyll fluorescence (SIFnet), *Biogeosci.*, 19, 1777–1793, doi:10.5194/bg-19-1777-2022.
32. Fitzmaurice, H., **A. J. Turner**, J. Kim, K. Chan, E. Delaria, C. Newman, P. J. Wooldridge, and R. C. Cohen (2022), Assessing vehicle fuel efficiency using a dense network of CO₂ observations, *Atmos. Chem. Phys.*, 22, 3891–3900, doi:10.5194/acp-22-3891-2022.
31. Wang, X., J. A. Biederman, J. F. Knowles, R. L. Scott, **A. J. Turner**, M. P. Dannenberg, P. Köhler, C. Frankenberg, M. E. Litvak, G. N. Flerchinger, B. E. Law, H. Kwon, S. C. Reed, W. J. Parton, G. A. Barron-Gafford, and W. K. Smith (2022), Satellite solar-induced chlorophyll fluorescence and near-infrared reflectance observations capture complementary aspects of dryland vegetation dynamics, *Remote Sens. Environ.*, 270, 112858–112869, doi:10.1016/j.rse.2021.112858.
30. **Turner, A. J.**, P. Köhler, T. S. Magney, C. Frankenberg, I. Fung, and R. C. Cohen (2021), Extreme events driving year-to-year differences in gross primary productivity across the US, *Biogeosci.*, 18, 6579–6588, doi:10.5194/bg-18-6579-2021.
29. Delaria, E. R., B. K. Place, **A. J. Turner**, Q. Zhu, X. Jin, and R. C. Cohen (2021), Development of a solar induced fluorescence-canopy conductance model and its application to stomatal reactive nitrogen deposition, *ACS Earth Space Chem.*, 5, 3414–3428, doi:10.1021/acsearthspacechem.1c00260.
28. Laughner, J. L., J. L. Neu, D. Schimel, P. O. Wennberg, K. Barsanti, K. Bowman, A. Chatterjee, B. Croes, H. Fitzmaurice, D. K. Henze, J. Kim, E. A. Kort, Z. Liu, K. Miyazaki, **A. J. Turner**, S. Anenberg, J. Avise, H. Cao, D. Crisp, J. de Gouw, A. Eldering, J. Fyfe, D. L. Goldberg, K. R. Gurney, S. Hasheminassab, F. Hopkins, C. E. Ivey, D. B. A. Jones, J. Liu, N. S. Lovenduski, R. V. Martin, G. A. McKinley, L. Ott, B. Poulter, M. Ru, S. P. Sander, N. Swart, Y. L. Yung, Z. Zeng, and KISS COVID-19 workshop team (2021), Societal shifts due to COVID-19 reveal large-scale complexities and feedbacks between atmospheric chemistry and climate change, *Proc. Natl. Acad. Sci.*, 118, doi:10.1073/pnas.2109481118.
27. Gensheimer, J., **A. J. Turner**, A. Shekhar, A. Wenzel, F. N. Keutsch, and J. Chen (2021), What are different measures of mobility changes telling us about emissions during the COVID-19 pandemic?, *J. Geophys. Res.*, 126, doi:10.1029/2021JD034664.
26. **Turner, A. J.**, J. Kim, H. Fitzmaurice, C. Newman, K. Worthington, K. Chan, P. J. Wooldridge, P. Köhler, C. Frankenberg, and R. C. Cohen (2020), Observed impacts of COVID-19 on urban CO₂ emissions, *Geophys. Res. Lett.*, 47, doi:10.1029/2020GL090037.

25. **Turner, A. J.**, P. Köhler, T. S. Magney, C. Frankenberg, I. Fung, and R. C. Cohen (2020), A double peak in the seasonality of California's photosynthesis as observed from space, *Biogeosci.*, *17*, 405–422, doi:10.5194/bg-17-405-2020.
24. Nguyen, N., **A. J. Turner**, Y. Yin, M. Prather, and C. Frankenberg (2020), Effects of chemical feedbacks on decadal methane emissions estimates, *Geophys. Res. Lett.*, *47*, doi:10.1029/2019GL085706.
23. **Turner, A. J.**[#], C. Frankenberg[#], and E. A. Kort[#] (2019), Interpreting contemporary trends in atmospheric methane, *Proc. Natl. Acad. Sci.*, *116*, 2805–2813, doi:10.1073/pnas.1814297116.
22. Cusworth, D. H., D. J. Jacob, J. X. Sheng, J. Benmergui, **A. J. Turner**, J. Brandman, L. White, and C. A. Randles (2018), Detecting high-emitting methane sources in oil/gas fields using satellite observations, *Atmos. Chem. Phys.*, *18*, 16885–16896, doi:10.5194/acp-18-16885-2018.
21. **Turner, A. J.**, I. Fung, V. Naik, L. W. Horowitz, and R. C. Cohen (2018), Modulation of hydroxyl variability by ENSO in the absence of external forcing, *Proc. Natl. Acad. Sci.*, *115*, 8931–8936, doi:10.1073/pnas.1807532115.
20. Sheng, J. X., D. J. Jacob, **A. J. Turner**, J. D. Maasakkers, J. Benmergui, A. A. Bloom, C. Arndt, R. Gautam, D. Zavala-Araiza, H. Boesch, and R. J. Parker (2018), 2010–2016 methane trends over Canada, the United States, and Mexico observed by the GOSAT satellite: contributions from different source sectors, *Atmos. Chem. Phys.*, *18*, 12257–12267, doi:10.5194/acp-18-12257-2018.
19. **Turner, A. J.**, D. J. Jacob, J. Benmergui, J. Brandman, L. White, and C. A. Randles (2018), Assessing the capability of different satellite observing configurations to resolve the distribution of methane emissions at kilometer scales, *Atmos. Chem. Phys.*, *18*, 8265–8278, doi:10.5194/acp-18-8265-2018.
18. Sheng, J. X., D. J. Jacob, **A. J. Turner**, J. D. Maasakkers, M. P. Sulprizio, A. A. Bloom, A. E. Andrews, and D. Wunch (2018), High-resolution inversion of methane emissions in the Southeast US using SEAC⁴RS aircraft observations of atmospheric methane: anthropogenic and wetlands sources, *Atmos. Chem. Phys.*, *18*, 6483–6491, doi:10.5194/acp-18-6483-2018.
17. Bloom, A. A., K. Bowman, M. Lee, **A. J. Turner**, R. Schroeder, J. R. Worden, R. J. Weidner, K. McDonald, and D. J. Jacob (2017), A global wetland methane emissions and uncertainty dataset for atmospheric chemical transport models (WetCHARTs version 1.0), *Geosci. Model Dev.*, *10*, 2141–2156, doi:10.5194/gmd-10-2141-2017.
16. Buchwitz, M., O. Schneising, M. Reuter, J. Heymann, S. Krautwurst, H. Bovensmann, J. P. Burrows, H. Boesch, R. J. Parker, P. Somkuti, R. G. Detmers, O. P. Hasekamp, I. Aben, A. Butz, C. Frankenberg, and **A. J. Turner** (2017), Satellite-derived methane hotspot emission estimates using a fast data-driven method, *Atmos. Chem. Phys.*, *17*, 5751–5774, doi:10.5194/acp-17-5751-2017.
15. **Turner, A. J.**, C. Frankenberg, P. O. Wennberg, and D. J. Jacob (2017), Ambiguity in the causes for decadal trends in atmospheric methane and hydroxyl, *Proc. Natl. Acad. Sci.*, *114*, 5367–5372, doi:10.1073/pnas.1616020114.
14. Tzompa-Sosa, Z. A., E. V. Fischer, E. Mahieu, B. Franco, C. A. Keller, **A. J. Turner**, D. Helmig, A. Fried, D. Richter, P. Weibring, J. Walega, T. I. Yacovitch, S. C. Herndon, D. R. Blake, F. Hase, J. Hannigan, S. Conway, K. Strong, and M. Schneider (2017), Revisiting global fossil fuel and biofuel emissions of ethane, *J. Geophys. Res.*, *122*, 2493–2512, doi:10.1002/2016JD025767.
13. Bader, W., B. Bovy, S. Conway, K. Strong, D. Smale, **A. J. Turner**, T. Blumenstock, C. Boone, M. C. Coen, A. Coulon, O. Garcia, D. W. T. Griffith, F. Hase, P. Hausmann, N. Jones, P. Krummel, I. Murata, I. Morino, H. Nakajima, S. O'Doherty, C. Paton-Walsh, J. Robinson, R. Sandrin, M. Schneider, C. Servais, R. Sussmann, and E. Mahieu (2017), The recent increase of atmospheric methane from 10 years of ground-based NDACC FTIR observations since 2005, *Atmos. Chem. Phys.*, *17*, 2255–2277, doi:10.5194/acp-17-2255-2017.
12. Maasakkers, J. D., D. J. Jacob, M. Sulprizio, **A. J. Turner**, M. Weitz, T. Wirth, C. Hight, M. DeFigueiredo, M. Desai, R. Schmeltz, L. Hockstad, A. A. Bloom, K. W. Bowman, S. Jeong, and M. L. Fischer (2016), A gridded national inventory of US methane emissions, *Environ. Sci. Technol.*, *50*, 13123–13133, doi:10.1021/acs.est.6b02878.
11. Jacob, D. J., **A. J. Turner**, J. D. Maasakkers, J. Sheng, K. Sun, X. Liu, K. Chance, I. Aben, J. McKeever, and C. Frankenberg (2016), Satellite observations of atmospheric methane and their application to constrain emissions, *Atmos. Chem. Phys.*, *16*, 14371–14396, doi:10.5194/acp-16-14371-2016.
10. **Turner, A. J.**, A. A. Shusterman, B. C. McDonald, V. Teige, R. A. Harley, and R. C. Cohen (2016), Network design for quantifying urban CO₂ emissions: assessing trade-offs between precision and network density, *Atmos. Chem. Phys.*, *16*, 13465–13475, doi:10.5194/acp-16-13465-2016.

9. Shusterman, A. A., V. Teige, **A. J. Turner**, C. Newman, J. Kim, and R. C. Cohen (2016), The BErkeley Atmospheric CO₂ Observation Network: Initial Evaluation, *Atmos. Chem. Phys.*, *16*, 13449–13463, doi:10.5194/acp-16-13449-2016.
8. Tan, Z., Q. Zhuang, D. K. Henze, C. Frankenberg, E. Dlugokencky, C. Sweeney, **A. J. Turner**, M. Sasakawa, and T. Machida (2016), Inverse modeling of pan-Arctic methane emissions at high spatial resolution: What can we learn from assimilating satellite retrievals and using different process-based wetland and lake biogeochemical models?, *Atmos. Chem. Phys.*, *16*, 12649–12666, doi:10.5194/acp-16-12649-2016.
7. Bousserez, N., D. K. Henze, B. Rooney, A. Perkins, K. J. Wecht, **A. J. Turner**, V. Natraj, and J. R. Worden (2016), Constraints on methane emissions in North America from future geostationary remote sensing measurements, *Atmos. Chem. Phys.*, *16*, 6175–6190, doi:10.5194/acp-16-6175-2016.
6. **Turner, A. J.**, D. J. Jacob, J. Benmergui, S. C. Wofsy, J. D. Maasakkers, A. Butz, O. Hasekamp, and S. C. Biraud (2016), A large increase in U.S. methane emissions over the past decade inferred from satellite data and surface observations, *Geophys. Res. Lett.*, *43*, doi:10.1002/2016GL067987.
5. Worden, J. R., **A. J. Turner**, A. Bloom, S. S. Kulawik, J. Liu, M. Lee, R. Weidner, K. Bowman, C. Frankenberg, R. J. Parker, and V. H. Payne (2015), Quantifying Lower Tropospheric Methane Concentrations Using Near-IR and Thermal IR Satellite Measurements: Comparison to the GEOS-Chem model, *Atmos. Meas. Tech.*, *8*, 3433–3445, doi:10.5194/amt-8-3433-2015.
4. **Turner, A. J.**, D. J. Jacob, K. J. Wecht, J. D. Maasakkers, E. Lundgren, A. E. Andrews, S. C. Biraud, H. Boesch, K. W. Bowman, N. M. Deutscher, M. K. Dubey, D. W. T. Griffith, F. Hase, A. Kuze, J. Notholt, H. Ohyama, R. Parker, V. H. Payne, R. Sussmann, C. Sweeney, V. A. Velazco, T. Warneke, P. O. Wennberg, and D. Wunch (2015), Estimating global and North American methane emissions with high spatial resolution using GOSAT satellite data, *Atmos. Chem. Phys.*, *15*, 7049–7069, doi:10.5194/acp-15-7049-2015.
3. **Turner, A. J.** and D. J. Jacob (2015), Balancing aggregation and smoothing errors in inverse models, *Atmos. Chem. Phys.*, *15*, 7039–7048, doi:10.5194/acp-15-7039-2015.
2. **Turner, A. J.**, A. M. Fiore, L. W. Horowitz, and M. Bauer (2013), Summertime cyclone frequencies over the Great Lakes Storm Track from 1860–2100: variability, trends, and association with ozone pollution, *Atmos. Chem. Phys.*, *13*, 565–578, doi:10.5194/acp-13-565-2013.
1. **Turner, A. J.**, D. K. Henze, R. V. Martin, and A. Hakami (2012), The spatial extent of source influences on modeled column concentrations of short-lived species, *Geophys. Res. Lett.*, *39*, L12806, doi:10.1029/2012GL051832.

INVITED SEMINARS

- 2026 MIT Program in Atmospheres, Oceans, and Climate (PAOC) Retreat
- 2026 Colorado State University
- 2026 University of Washington (Department of Statistics)
- 2025 University of Washington (Chemical Oceanography)
- 2025 Massachusetts Institute of Technology (MIT)
- 2025 University of California at Berkeley
- 2025 University of Southern California (USC)
- 2024 Massachusetts Institute of Technology (MIT)
- 2024 NASA Ames Research Center
- 2023 California Institute of Technology (Caltech)
- 2023 University of Utah
- 2022 NASA Ames Research Center
- 2022 MIT Frontiers in Atmospheric Chemistry Seminar Series (FACSS)
- 2022 Stanford University
- 2022 NASA Goddard Space Flight Center
- 2022 NASA Ames Research Center
- 2022 University of Rochester
- 2021 Environment and Climate Change Canada
- 2021 University of Toronto
- 2021 UC Berkeley Climate and Impacts Group
- 2021 Imperial College London
- 2020 NASA Jet Propulsion Laboratory (JPL)
- 2020 University of California at Berkeley
- 2020 University of Washington
- 2019 Bay Area Air Quality Management District (BAAQMD)

2019 OneNOAA Science Seminar
 2018 Stanford University
 2018 University of California at Berkeley
 2018 Technical University of Munich (TUM), Germany
 2018 NASA Ames Research Center
 2018 NOAA Geophysical Fluid Dynamics Laboratory (GFDL)
 2018 University of Washington
 2017 Japanese Aerospace Exploration Agency (JAXA)
 2017 California Institute of Technology (Caltech)

ORAL CONFERENCE PRESENTATIONS ([†]INVITED)

2026[†] OCO Science and Applications Center Workshop, Caltech, Pasadena, CA
 2026[†] Workshop on Machine Learning for Atmospheric Chemistry & Aerosols, University of Reading, UK, virtual
 2026[†] Schmidt Sciences' Virtual Institute for the Carbon Cycle (VICC) Phase 2 Scoping Workshop, New York, NY
 2026[†] Methane Action for People & Planet, European Commission Joint Research Centre, Ispra, Italy
 2025[†] AGU Fall Meeting: Session on Methodologies and Datasets for Tracking GHG Emissions, New Orleans, LA
 2025[†] Schmidt Sciences' Virtual Earth System Research Institute (VESRI) Virtual Symposia, virtual
 2025[†] UW Program on Climate Change summer institute at Friday Harbor Labs, San Juan Islands, WA
 2025[†] Telluride Science Workshop: Mapping Urban Air, Telluride, CO
 2025[†] Urban greenhouse gas seminar series, virtual
 2024[†] AGU Fall Meeting: Session on Tropospheric Oxidative Capacity, Washington, DC
 2024[†] AGU Fall Meeting: Session on Vintage Atmospheric Chemistry, Washington, DC
 2024[†] EPA Methane Inverse Modeling Technical Workshop, Durham, NC
 2024[†] Schmidt Sciences' Cross-VESRI Convening, University of Cambridge, Cambridge, UK
 2024[†] Summer School for Inverse Modeling of Greenhouse Gases, Colorado State University, Fort Collins, CO
 2023[†] AGU Fall Meeting: Session on Data-driven Methods, San Francisco, CA
 2023[†] Gordon Conference on Atmospheric Chemistry, Sunday River, ME
 2023[†] Telluride Science Workshop: Mapping Urban Air, Telluride, CO
 2023[†] Canadian Society for Chemistry (CSC) Symposium on Atmospheric Organics, Vancouver, Canada
 2022 AGU Fall Meeting: Session on Ice Cores, Chicago, IL
 2022 OCO-2 Science Team Meeting, virtual
 2021[†] Telluride Science Workshop: Mapping Urban Air, Telluride, CO
 2020[†] AGU Fall Meeting: James R. Holton Award talk in the Frontiers of Atmospheric Science session, virtual
 2020[†] AGU Fall Meeting: Union Session on COVID-19 in the Earth system, virtual
 2020 AGU Fall Meeting: Session on Solar-Induced Chlorophyll Fluorescence, virtual
 2020 Air Sensors International Conference (ASIC) Fall series, virtual
 2020[†] COVID-19: Identifying Unique Opportunities for Earth Science, Keck Institute for Space Studies, virtual
 2019 AGU Fall Meeting: Session on Solar-Induced Chlorophyll Fluorescence, San Francisco, CA
 2019[†] CO₂-Urban Synthesis and Analysis (CO₂-USA) Workshop, Boston University
 2019[†] Frontiers of Atmospheric Science and Chemistry (FASCINATE 2019), NCAR, Boulder, CO
 2019[†] Global Air Quality Sensing Forum, Berkeley, CA
 2017 AGU Fall Meeting: Session on Remote-sensing of CO₂ and CH₄, New Orleans, LA
 2017[†] UN Climate Change Conference (COP23), Bonn, Germany
 2017 Atmospheric Chemistry Colloquium for Emerging Senior Scientists (ACCESS) XIV, Brookhaven National Lab
 2017[†] DOE Computational Science Graduate Fellowship Program Review, Washington, DC

TEACHING EXPERIENCE

University of Washington: *Received the annual department teaching award in 2022, 2023, & 2026*

- ATMOS 310 (Numerical Methods): Autumn 2026
- ATMOS 340 (Atmospheric Thermodynamics): Winter 2023, 2026
- ATMOS 358 (Atmospheric Chemistry): Spring 2021, 2022, 2023, 2024, 2025, 2026
- ATMOS 501 (Atmospheric Physics & Chemistry): Autumn 2021, 2022, 2023, 2024, 2025
- ATMOS 532 (Atmospheric Radiation): Winter 2022
- Climate & Environmental Justice Faculty Development Workshop: Winter 2022, 2023
- Guest lecturer in ATMOS 220 (Exploring Atmospheric Science): Winter 2022, Spring 2023, 2026
- Guest lecturer in ATMOS 558 (Carbon Cycle): Winter 2022, 2023

MENTORING

Postdocs ($n = 2$)

- Dr. Hikari Murayama, 2026 – *present*
- Dr. Tai-Long He, 2022 – 2024. Next position: Postdoc at Harvard

PhD students ($n = 9$)

- Jacob Chalif, 2026 – *present*
- Toireasa-Marie O'Rourke, 2026 – *present*
- Sydney Vernon, 2026 – *present*
- Lucy Zimmerman, 2026 – *present*
- Iana Ferguson, 2024 – *present*
- Eliot Kim, 2024 – *present*
- Eric Mei, 2023 – *present*
- James Yoon, 2022 – *present*
- Dr. Nikhil Dadheech, 2021 – 2026 PhD. Next position: Applied Scientist at Amazon

Masters students ($n = 1$)

- Johannes Gensheimer, 2020 – 2021 MS (at TUM). Next position: PhD student at MPI Jena

Postbac Researchers ($n = 2$)

- Isabella Showman, 2026 – *present*
- Sydney Vernon, 2025 – 2026. Next position: PhD student at U. Washington

Undergraduates ($n = 9$)

- Srinidhi Muralidharan, 2026 – *present*
- Isabella Showman, 2025 – 2026. Next position: Postbac researcher at U. Washington
- Coco Lipe, 2024 – 2026. Next position: Air quality consultant at Ramboll
- Laura Pong, 2023 – 2026. Next position: PhD student at U. Toronto
- Lauren Yarrington, 2023 – 2026.
- Alyssa Tou, 2024 – 2025. Next position: Air quality consultant at Ramboll
- Max Taniguchi-King, Summer 2024. Next position: Software engineer at Data Science & Environment
- Simon Zhang, 2022 – 2023. Next position: PhD student at U. Minnesota
- Ryan Boyd, 2021 – 2023. Next position: PhD student at Princeton

SERVICE ACTIVITIES

Departmental:

- UW ATMOS computing committee member, 2024 – *present*.
- UW ATMOS strategic planning committee member, 2023 – *present*.
- UW ATMOS graduate curriculum committee member, 2021 – *present*.
- UW ATMOS assistant computing director search committee member, 2024.
- UW ATMOS committee on graduate studies, 2023 – 2025.
- UW ATMOS faculty search committee member, 2023 – 2024.
- UW ATMOS teaching schedule committee member, 2022 – 2023.

College/University:

- UW AI@UW advisory board member, 2026 – *present*.
- UW college scholarship committee member, 2024 – 2026.
- UW Program on Climate Change (PCC) postdoctoral fellowship reviewer, 2026.
- UW task force to develop an AI minor, 2026.
- UW Atmospheric and Climate Science department chair search committee member, 2025 – 2026.
- UW university research resilience & transformation advisory committee, 2025 – 2026.
- UW co-director of “Computing for the Environment” initiative, 2022 – 2025.
- UW facilitator of climate & environmental justice faculty development workshop, 2022 – 2023.
- UW faculty senator, 2021 – 2023.

National:

Science Teams

- Science team member for the NASA STRIVE satellite mission, 2024 – *present*.
- Science team member for the NASA Carbon-I satellite mission concept, 2024 – 2026.

White papers

- Co-author, white paper for National Academies Earth Science decadal survey: “*Observationally constraining the spatio-temporal abundance, trends, and variability of the hydroxyl radical (OH) and its drivers*” (Anderson et al., 2026).

- Co-author, white paper for National Academies Earth Science decadal survey: *“High Spatial Resolution Greenhouse Gas Observations: Closing Critical Data Gaps in the Global Carbon Cycle”* (Thorpe et al., 2026).
- Co-author, white paper for National Academies Earth Science decadal survey: *“Observationally Constraining Magnitudes and Trends of Urban CO₂ and CH₄ Emissions”* (Ahn et al., 2026).
- Co-author, white paper for NASA Atmosphere community: *“GHG Implications”* (Laughner et al., 2026).
- Co-author, white paper from Microsoft Research workshop on Urban Futures: *“Why all cities should have ‘Clean Air as a City Service’”*, (Lauter et al., 2020).

Other

- Committee member, National Academies Panel on *“Atmospheric Methane Removal”*, 2023 – 2024.
- Co-organizer, Telluride Science Research Conference on *“Mapping Urban Air: Linking Observations and Processes”*, 2021 & 2025.
- Team Lead, Keck Institute for Space Studies workshop on *“COVID-19 and the Earth System”*, 2021.

International:

- Project lead for FETCH₄, an international collaboration of scientists focused on improving our understanding of the past and modern methane cycle (28 institutions across 8 countries), 2023 – *present*.
- External referee for the Dutch Research Council’s Stevin Prize, one of the Netherlands’ highest honors, 2025.
- Lead chapter author, Japanese National Institute for Environmental Studies report: *“A guidebook on the use of satellite greenhouse gases observation data to evaluate and improve greenhouse gas emission inventories”*, 2018.
- Co-chair, session at AGU Fall Meeting: 2019, 2020, 2025.
- Guest editor for *PNAS*.
- Peer reviewer and/or review panel member for: *NSF*, *NASA*, *NOAA*, *DFG (Germany)*, *Schmidt Sciences*, and *Stanford Sustainability Accelerator*.
- Regular peer reviewer for: *Science*, *Nature*, *PNAS*, *Science Advances*, *Nature Geoscience*, *Geophysical Research Letters*, *Atmospheric Chemistry and Physics*, *Biogeosciences*, *Atmospheric Measurement Techniques*, *Geoscientific Model Development*, *Environmental Science & Technology*, *Earth’s Future*, and *Journal of Geophysical Research*.